

Operating Systems: Process Scheduling

Introduction to Process Scheduling

Process scheduling is the activity of the operating system that decides which process in the ready queue gets access to the CPU next. Since a single CPU core can execute only one process at a time, the scheduler must choose efficiently to keep the system responsive and to make good use of available resources.

The main goals of a scheduling algorithm include maximizing CPU utilization, maximizing throughput, minimizing turnaround time, minimizing waiting time, and minimizing response time. Different scheduling algorithms make different tradeoffs between these goals depending on the type of system being designed.

First-Come First-Served (FCFS)

FCFS is the simplest scheduling algorithm. Processes are executed in the order they arrive in the ready queue, much like a queue at a ticket counter. While easy to implement, FCFS suffers from the convoy effect, where a long process at the front of the queue causes all shorter processes behind it to wait an unfairly long time.

Shortest Job First (SJF)

SJF selects the process with the smallest estimated execution time next. This algorithm is provably optimal for minimizing average waiting time, but it requires knowing (or estimating) the burst time of each process in advance, which is not always practical in real systems.

A preemptive version of SJF, called Shortest Remaining Time First (SRTF), allows a newly arrived process with a shorter burst time to interrupt the currently running process.

Round Robin (RR)

Round Robin assigns each process a small fixed unit of CPU time, called a time quantum, and cycles through the ready queue. If a process does not finish within its quantum, it is placed at the back of the queue and the next process runs. Round Robin is widely used in time-sharing systems because it guarantees fairness and reasonable response time for interactive processes.

Choosing the time quantum involves a tradeoff: a very small quantum increases context-switching overhead, while a very large quantum makes Round Robin behave like FCFS.

Priority Scheduling

In priority scheduling, each process is assigned a priority, and the CPU is allocated to the process with the highest priority. A significant problem with this approach is starvation, where low-priority processes

may never execute if higher-priority processes keep arriving. This is commonly solved using aging, where a process's priority gradually increases the longer it waits in the queue.